

Touchable Cooled Graphics: Midair 3D Image with Noncontact Cooling Feedback using Ultrasound-Driven Mist Vaporization

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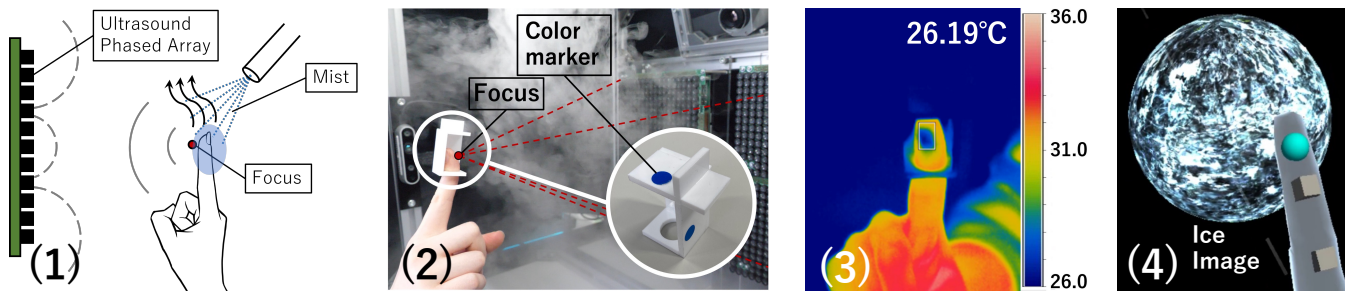


Figure 1: (1) Presentation of cooling sensation to a fingertip using ultrasound-driven mist vaporization. (2) Presenting cooling sensation with the system. (3) Thermal image indicating temperature decrease due to cold stimulation. (4) User's point of view in a demonstration.

ABSTRACT

Adding tactile feedback to a midair image realizes immersive mixed reality contents. In this study, we develop a midair 3D image with a noncontact cooling sensation using ultrasound. In this system, users can feel a cooling sensation when touching the 3D image with their bare hands. The noncontact cooling sensation is rapidly displayed by ultrasound-driven mist vaporization. In the previous ultrasound haptic-optic display, only mechanical tactile feedback e.g. vibration has been used. The cooling sensation can extend the displayable material texture of the ultrasound haptic system. In the demo, we present a 3D image of ice. Participants can touch the image freely and feel its realistic cooling sensation.

CCS CONCEPTS

• Human-centered computing → Haptic devices.

KEYWORDS

cooling sensation, haptic display, midair haptics, focused ultrasound

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SA '22 Emerging Technologies, December 06-09, 2022, Daegu, Republic of Korea

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ACM ISBN 978-1-4503-9472-7/22/12.

<https://doi.org/10.1145/3550471.3558402>

ACM Reference Format:

Hanaho Motoyama, Tao Morisaki, Masahiro Fujiwara, Yasutoshi Makino, and Hiroyuki Shinoda. 2022. Touchable Cooled Graphics: Midair 3D Image with Noncontact Cooling Feedback using Ultrasound-Driven Mist Vaporization. In *SIGGRAPH Asia 2022 Emerging Technologies (SA '22 Emerging Technologies)*, December 06-09, 2022. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3550471.3558402>

1 INTRODUCTION

Tactile feedback can enhance the sense of immersion into computer games and VR space.

To present tactile stimulus without restricting the user's motion, noncontact tactile presentation using airborne ultrasound phased array (AUPA) has been proposed [Carter et al. 2013; Hoshi et al. 2010]. AUPA is an array of ultrasonic transducers which can create an ultrasound focal point in midair by controlling the phase and amplitude of each transducer. At the focal point, a noncontact force called acoustic radiation force is generated, which delivers a noncontact tactile stimulus onto human skin.

By using AUPA, the touchable midair image can be realized. Since the noncontact tactile stimulus does not interfere with the user's vision, AUPA is particularly suitable for adding tactile sensation to midair 3D images. Many previous studies proposed the midair haptic-optic system using ultrasound and enriched the tactile experience in the midair image [Morisaki et al. 2021].

However, temperature information, which is an important sense in tactile perception, has been lacking in the haptic-optic system

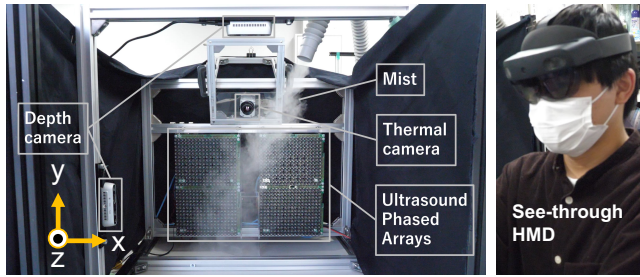


Figure 2: System configuration

using ultrasound. This lack limits the presentable material in the system. Humans use temperature information to distinguish the material of touched objects. For example, they can distinguish between metal and wood at room temperature by feeling their thermal conductivity.

In this study, we propose a midair haptic-optic system providing not only mechanical tactile stimulus (vibration, pressure) but also cooling sensation via focused ultrasound (Fig. 1). When users touch a presented midair 3D image, they can perceive mechanical tactile feedback and a cooling sensation. The cooling sensation was displayed using ultrasound-driven mist vaporization (Fig. 1-(1)). When injecting a mist-flow to the ultrasound focus, the mist volatilizes at high speed and the part of human skin at the focus position is cooled (Fig. 1-(2)). The mist vaporization method [Nakajima et al. 2021] can present a cooling sensation with pinpoint and at high speed compared to other noncontact cooling such as air flow [Kamigaki et al. 2020; Xu et al. 2019]. Therefore, this system can cool only the contact area between the 3D image and the user's finger in real-time.

In the demonstration, as a primitive cold object, a 3D image of ice is presented (Fig. 1-(4)). Participants can touch the 3D image of ice freely and feel a realistic cold sensation with their bare finger.

2 TOUCHABLE COOLED MIDAIR IMAGE

2.1 Hardware Properties

Our cooled midair image system (Fig. 2) consists of four AUPAs, a mist generator, two depth cameras (Intel Real Sense D435), and a see-through head-mounted display (Microsoft HoloLens2). The AUPAs were used to present ultrasound focus. One unit of AUPA contains 249 ultrasound transducers driving at 40 kHz [Suzuki et al. 2021].

2.2 Displaying Cooling Sensation

This system tracks the user's fingertip position and presents a cooling sensation to it when the fingertip touches the presented 3D image. Moreover, by modulating the sound pressure at the focus, the system can simultaneously present a cooling sensation and vibrotactile stimulation.

Fingertip position tracking is performed using two cameras. To measure the fingertip position robustly under the influence of mist flow, the two color markers were added to the fingertip (Fig. 1-(2)). One color marker is placed above the fingertip and the other marker

is placed on the side. The side camera detects the z-y position of the side marker, and the top camera detects the x position of the top marker. From these data, the 3D position of the fingertip is finally obtained. When the fingertip touches the presented 3D image, an ultrasound focus is presented to the measured 3D position. This focus volatilizes the mist and presents a cold sensation to the fingertips at fast. The mist is constantly being presented to the user's hands during the demonstration. We preliminarily confirmed that this system can lower the surface temperature of the finger by 1.2°C in 0.5 s (Fig.1-(3)).

3 DEMO AND POTENTIAL APPLICATION

In the demonstration, participants can feel a realistic cold sensation when they touch the 3D images of ice and other cold objects. In the authors' subjective view, when touching the image of ice, we felt the sensation of touching real cold ice with high reality.

In the future, we reproduce the change in skin temperature when touching various real objects. The temporal pattern of the skin temperature could be controlled by changing the sound pressure at an ultrasound focal point. We believe that skin temperature control realizes the reproduction of tactile feelings of various real objects e.g. ice, metal, and wood in the ultrasound haptic system.

ACKNOWLEDGMENTS

This work is supported in part by JST CREST JPMJCR18A2.

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